

Oral Examination of Advanced Power Electronics and Applications 2026

There will be an oral examination for 2nd semester (EPSH2, PED2, HWPS2 and APEL2) students. Every student will have to draw a question number.

Preparation time: 15-18 min preparation time (with full access to computers, literature, and other resources).

Oral Exam: Approx. 15- min oral examination. Except the question paper no other resource is permitted during the oral examination.

Here is a list of questions for the oral exam. One question will be selected randomly. Emphasis will be given to the logical analysis and presentation of the solution on blackboard/whiteboard. Computer is NOT allowed during the exam.

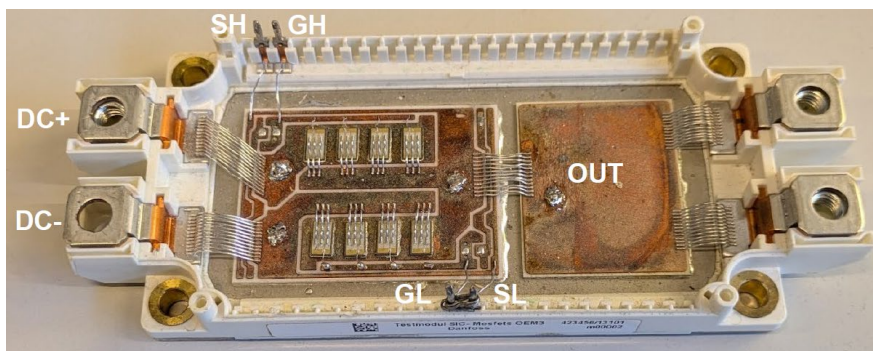
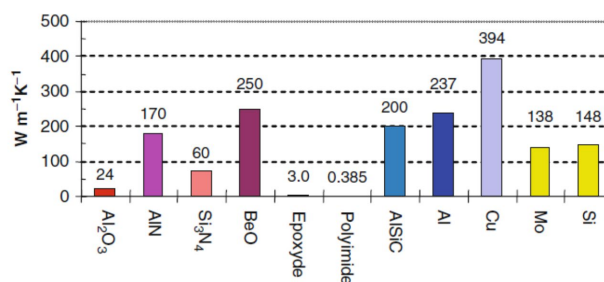
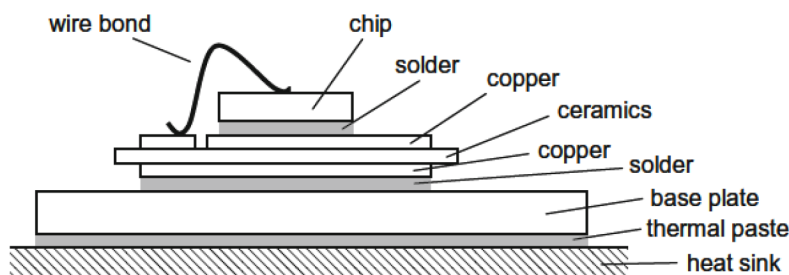
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Q1. Power modules and their properties

Due to finite feasible chip size and their current limit, several chips are packaged into a power module, which should ensure both good electrical and thermal performance.

- Explain the importance of the temperature dependency of the static characteristics (i-v curves) of power chip devices (i.e. MOSFET or IGBT), and whether this makes them more or less suitable for paralleling.
- Give examples of the origin of electrical circuit parasitic elements in a power module, and based on a circuit diagram describe how they can influence the switching performance of the power module.
- Explain the concept of an auxiliary source connection (also known as Kelvin-source connection), and show how it is physically implemented on a semiconductor device.
- Discuss how the material layer structure and properties of a power module gives rise to a tradeoff between thermal performance and parasitic capacitance.



Q2. [CSC]

[Ned Mohan, 6-18] In the three phase inverter of Fig. 6-33a, $V_{LL}=460$ V at 60 Hz, $E=550$ V, and $L_s=0.5$ mH. Assume L_d is very large, to yield $i_d(t)=I_d$. Calculate α and γ if the power flow is 55 kW. Explain the operation of such converters.

- Describe the operation of Graetz bridge converters. What are firing angle, and commutation overlap? Explain these angles with the help of a diagram for the rectifier mode.
- How can we change the operating mode from rectifier to inverter?
- What factors affect the commutation overlap?
- What is commutation margin? Why is it necessary?
- Explain the harmonics generated by a 6-pulse and 12 pulse LCC. Draw the ideal ac current waveform and dc voltage waveforms, used in the computation of the characteristic harmonics.

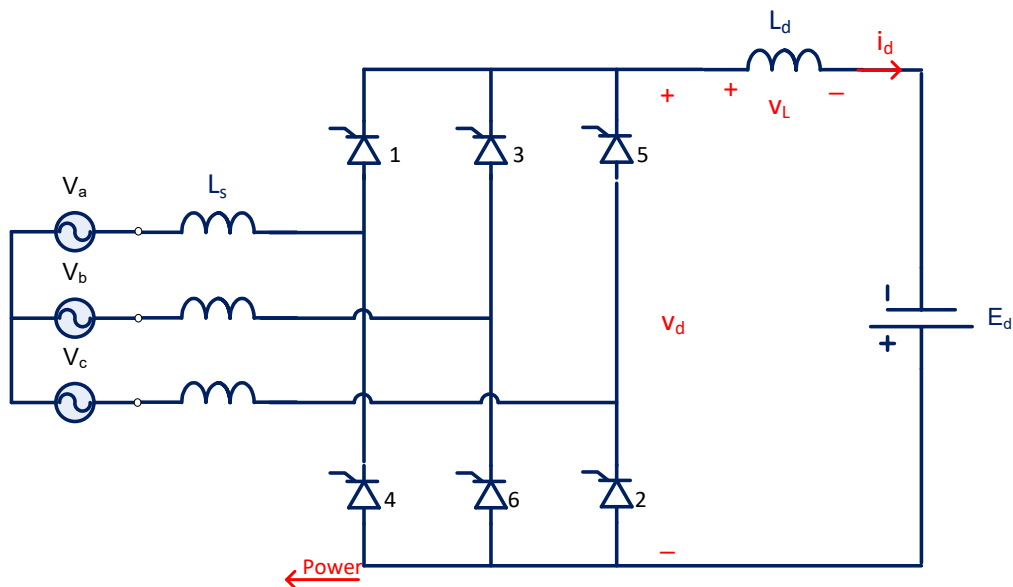


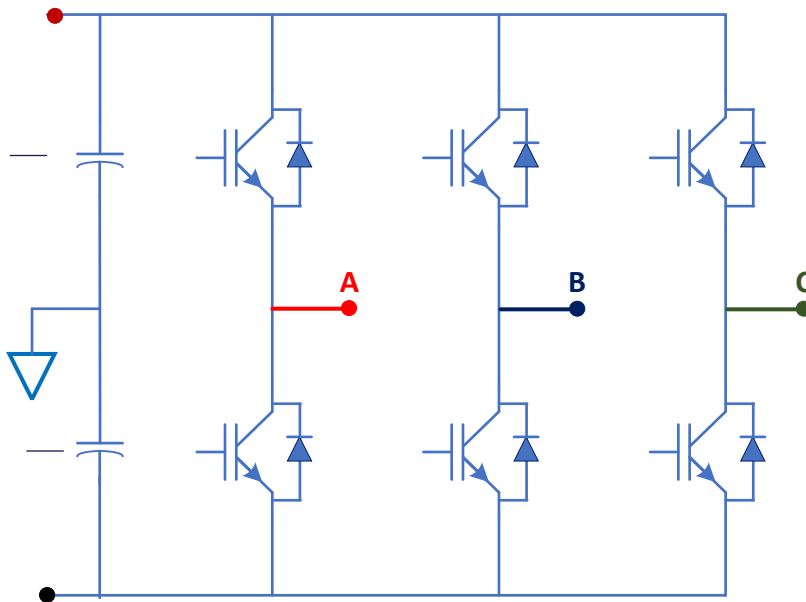
Fig 6-33 Thyristor inverter with a dc voltage source

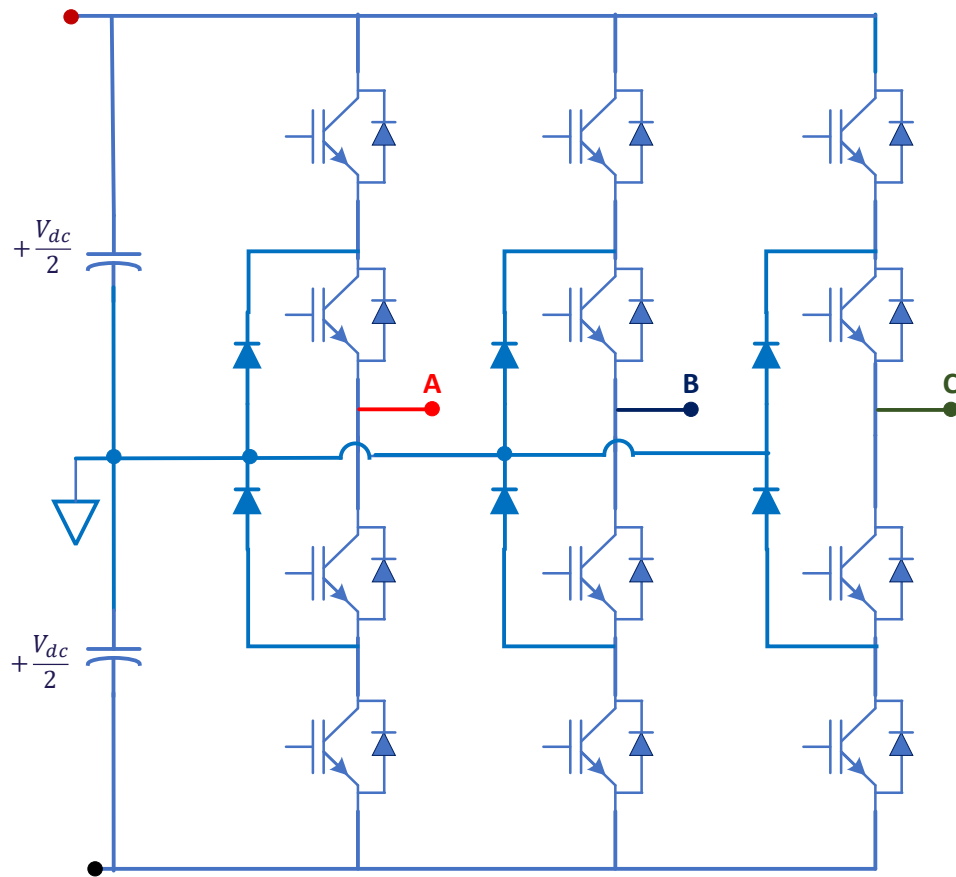
Ref. Power Electronics by Mohan, Undeland and Robbins

Q3. [VSC]

Consider a VSC-HVDC system using 2-level VSC.

- Compare a 2-level VSC with a 3-level NPC converter. Comment upon their pros and cons including harmonic spectrum
- What is the difference between a VSC and a CSC? Comment upon (i) switch type, (ii) filter requirements, and (iii) advantage/disadvantage.
- Describe the current control of a VSC using the dq transformations in the synchronously rotating reference frame. Explain with the help of a block diagram.
- Expand the block diagram drawn in (c) to describe the control of active and reactive power during the grid connected operation of a VSC.
- How do we control a VSC as (i) STATCOM and as (ii) HVDC converter.

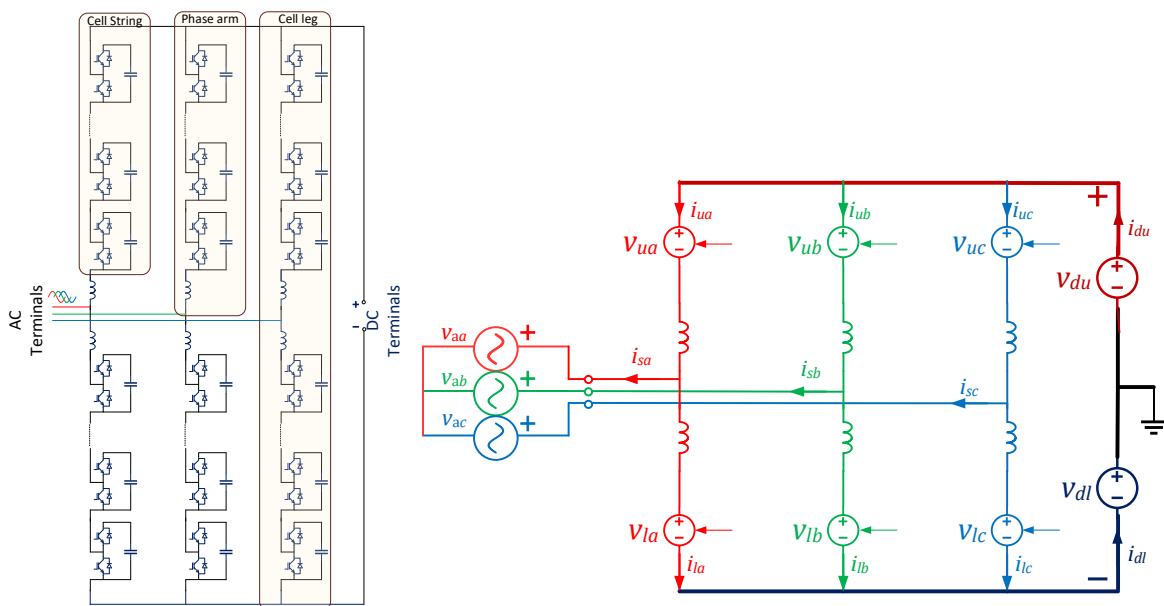




Q4. [MMC]

- Describe the basic topology and operation of MMC? What are the advantages and disadvantages? Why MMC needs arm inductor which is not present in other topologies?
- Describe the relationship between the arm current, dc current and the output ac current. How are they related with the active and reactive power output of the converter.
- Similarly describe the relationship between the arm voltages, dc voltage, output ac voltage and insertion indices.
- Using the relations in steps (b) and (c), describe the equations for controlling the output ac current and the circulating dc current.
- Why do we need to balance the capacitor voltages? How is it achieved?

Explain with the help of the circuit and equivalent circuit below.



Q5. [Soft switching and resonant converters DC/DC]

- Give three arguments why using soft switching is used?
- What are the basic resonant circuit types?

Given the SLR converter shown below:

- What components determine the resonant frequency?
- What is the duty-cycle of T_+ and T_- ?
- How is power dissipation in R controlled?
- Can there be used a transformer in the (SLR) converter?

Series Load Resonant (SLR) Converter

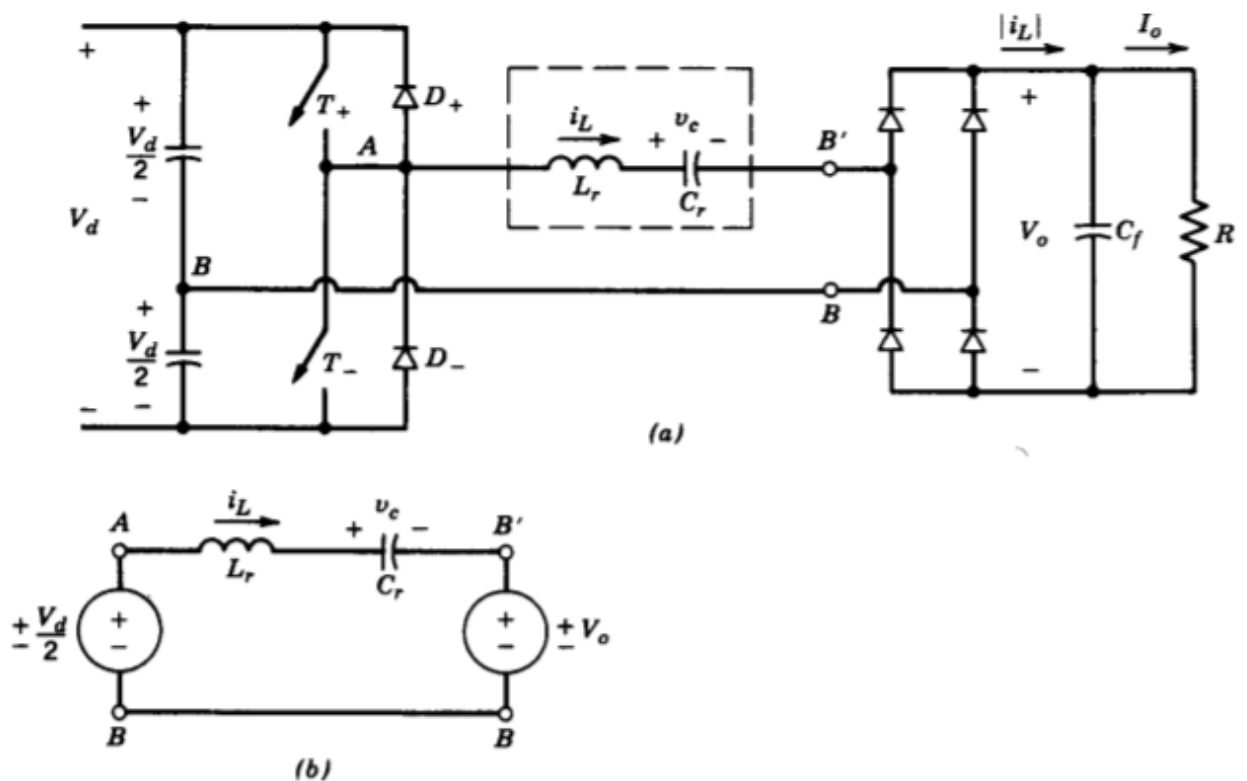


Figure 9-10 SLR dc–dc converter: (a) half-bridge; (b) equivalent circuit.

Q6. [Load-resonant and zero volt and current switched converters and applications]

- Give three examples of applications DC-to-AC resonant converters
- A furnace resonant converter is shown below, explain the operational behaviour including: 1. What is the duty-cycle of S1 and S2 given the control equation shown B3. 2. What determines the resonant frequency.
- What is the peak voltage of the switches S1 and S2 in steady state operation?
- In steady state what is the DC current value of inductor L_d as function of R ?

